

On March 15, 2013, the two funds merged. Janus Worldwide, the bigger fund with the longer history, ceased to exist and was absorbed by Janus Global Research, the much smaller fund with the shorter history. Why?

Panel B of Figure 16.3 plots the year-on-year returns of Janus Worldwide and Janus Global Research from February 2005 to January 2012. Janus Global Research has trounced its much bigger cousin. Over the sample, Janus Global Research averaged 9.9% while Janus Worldwide returned 5.0%. Since both funds have approximately the same standard deviation, at just under 20%, Janus Global Research's Sharpe ratio (see chapter 2) was double that of Janus Worldwide.

After the merger, the combined fund inherited only the superior performance of Janus Global Research. Janus Worldwide ceased to exist, and its inferior performance disappeared from the Janus fund family.

Killing or merging poorly performing funds is how Janus's marketing team can state with a straight face, "100 percent of Janus equity funds have beaten their benchmarks since inception."²⁴ *Survivorship bias* is the tendency of the worst funds, like Janus Worldwide, to vanish through mergers or dissolution. The result is that the universe of funds reporting returns presents far too rosy a picture. This is accomplished by leaving out the returns of euthanized funds.

How Large Is the Survivorship Bias?

Since all 40-Act funds are required to report their performance to the SEC, we observe the full universe of funds. We measure the effect of survivorship bias by comparing the returns of live funds, like Janus Global Research, to those that die or merge, like Janus Worldwide. The first large-scale mutual fund database without survivorship bias was put together by Mark Carhart when he was a PhD student at the University of Chicago. His data set, with updates and improvements, has been extensively used since. Carhart graduated in 1995, took a job as a finance professor at the University of Southern California, and shortly afterward was lured away to Goldman Sachs Asset Management. He ended up co-managing one of the largest quant hedge funds in the mid-2000s (which later dissolved after the financial crisis; see chapter 17).

Using only the live funds overstates mutual fund returns by 1% to 2% compared to a universe that includes both live and dead funds.²⁵ But the difference between live funds and the discontinued funds taken separately is about 4%.

²⁴Cruz, H., "Don't Disregard Past Performance When Choosing a Fund: Investment Advertisements Should Still Get a Critical Look," *Chicago Tribune*, May 20, 2010.

²⁵This is shown in Elton, Gruber, and Blake (1996) and Carhart's (1997) seminal paper. See also Brown et al. (1992), Carpenter and Lynch (1999), and Carhart et al. (2002). A subtle point is made by Linnainmaa (2013), who notes that while the group of mutual funds that die have lower returns than surviving funds, there can be a positive bias resulting from survivorship bias if fund-level data is used to compute mutual fund alphas, as some of the studies mentioned in Section 3.2, if the chance of a fund dying increases when a fund's alpha is low. There is no Linnainmaa survivorship bias for studies tracking all mutual funds, some of which die and some of which survive, at a point in time. Linnainmaa finds